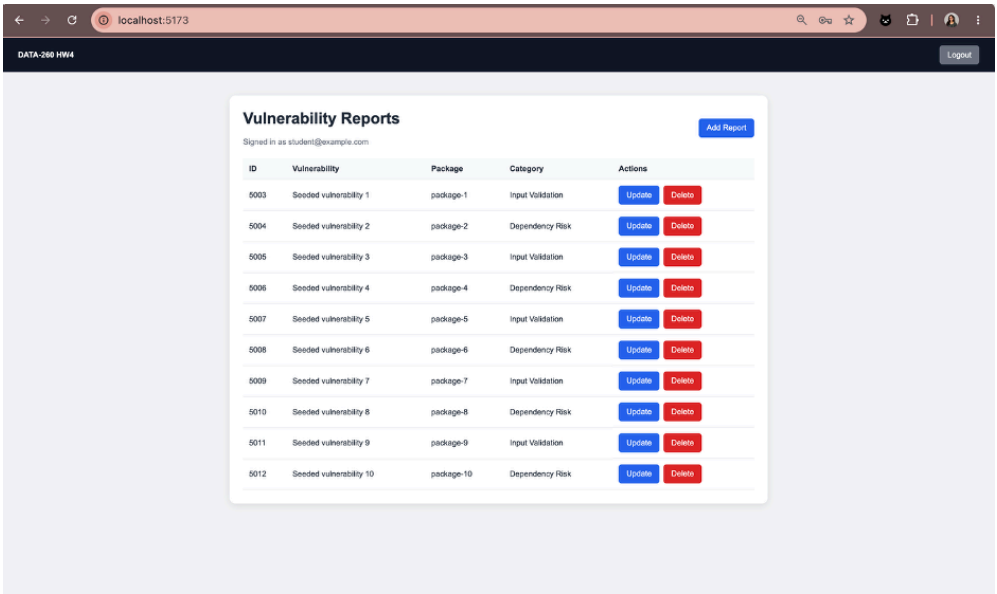
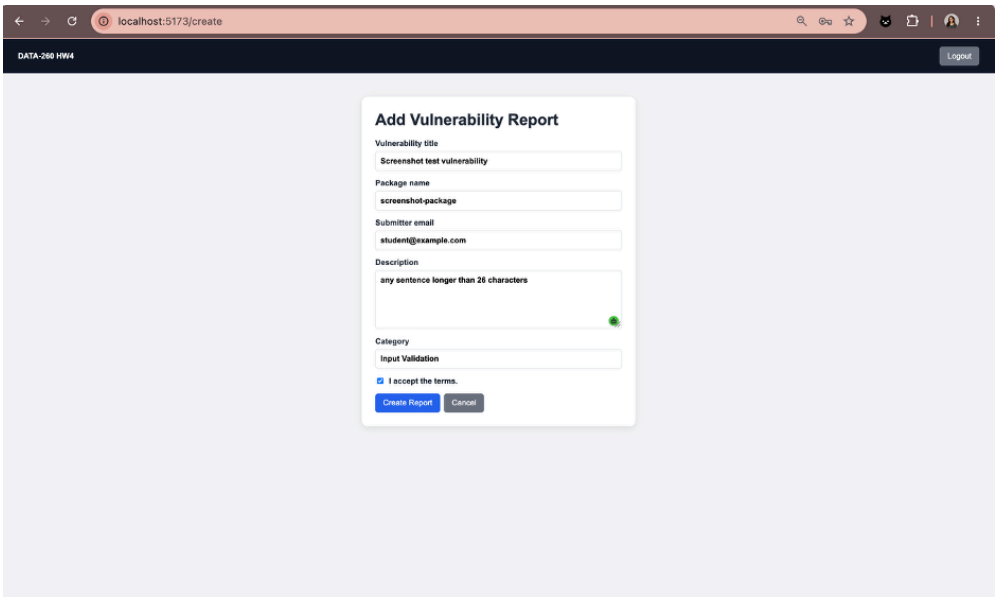


Additional Evidence Screenshots

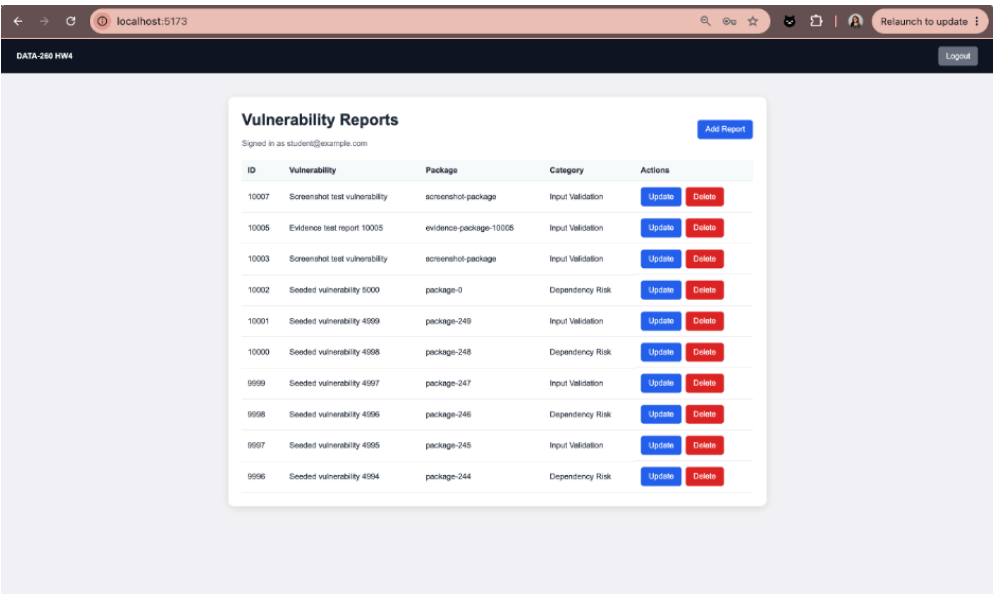
Frontend



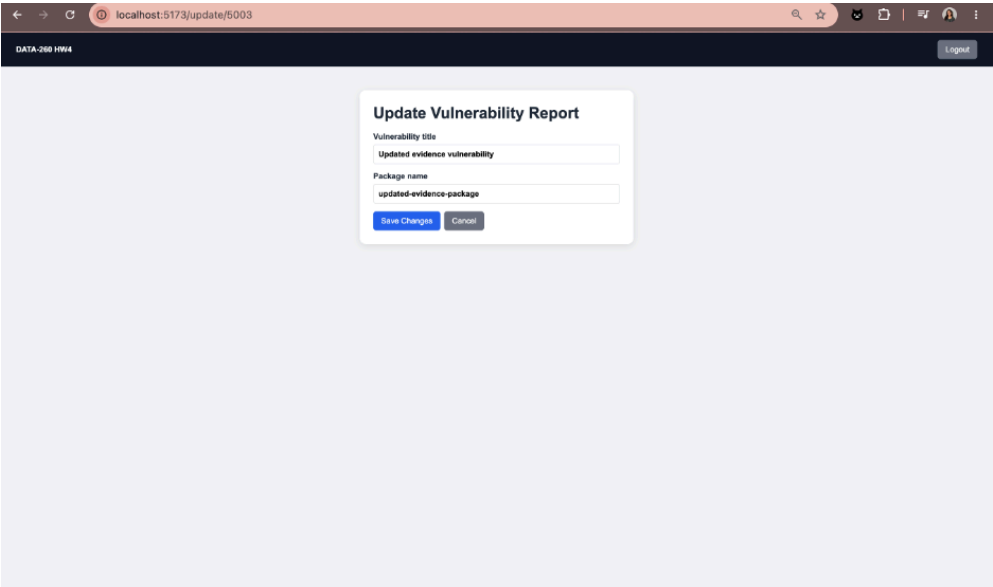
Login and dashboard



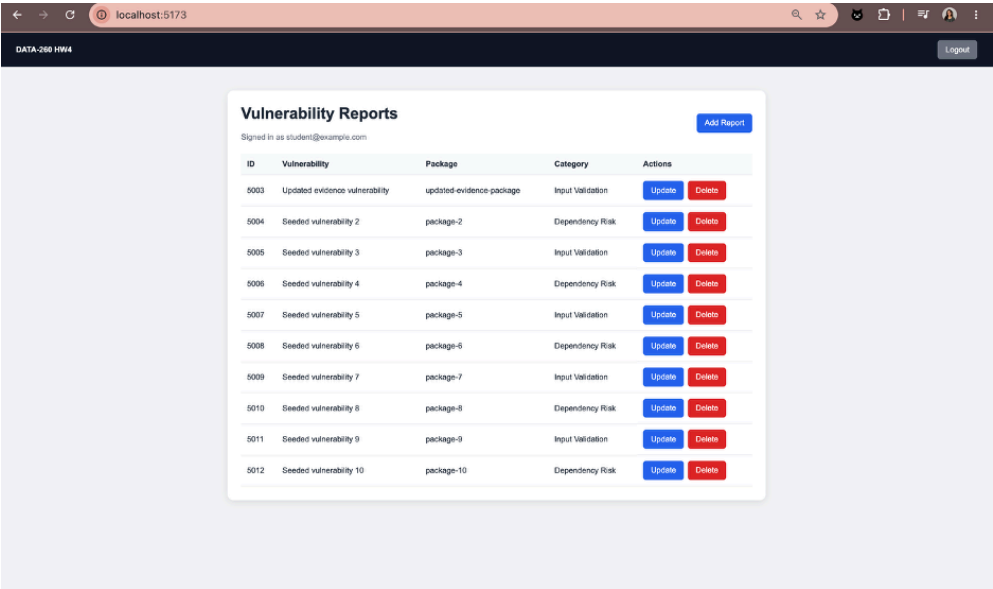
Create report form



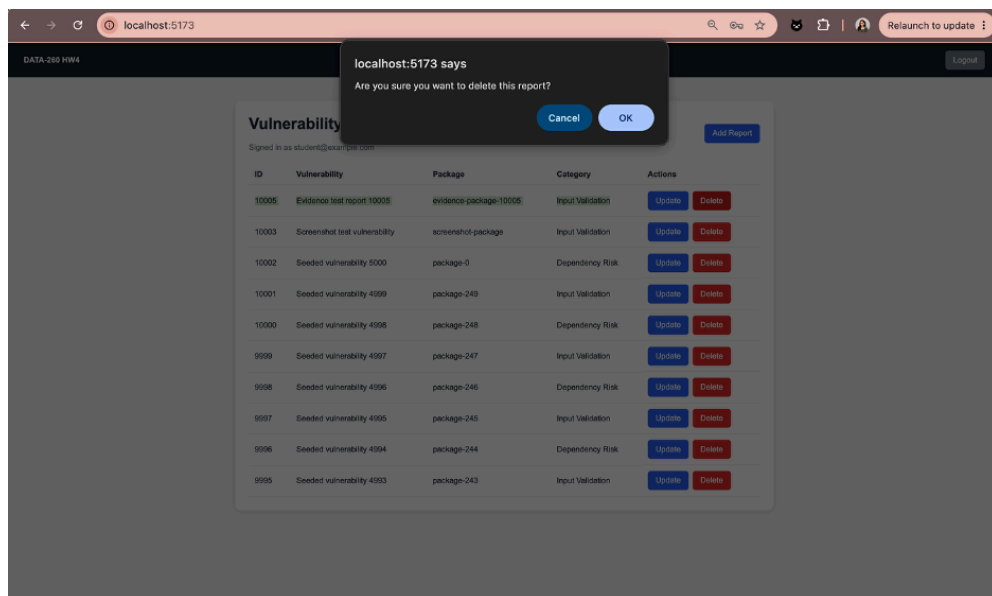
Create report result



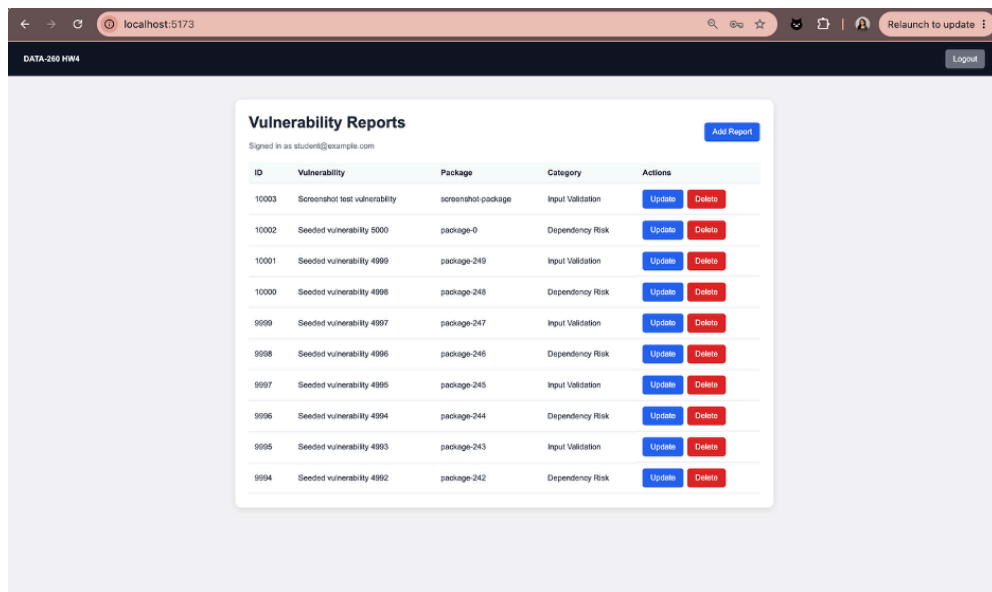
Update report form



Updated report result



Delete confirmation



Dashboard after deletion

API CRUD Operations

```
(base) sruti@Ashika-MacBook-Pro data260-1984 % cd ~/data260-1984

curl -s -u /tmp/hmcteachlee \
-H "Content-Type: application/json" \
-d '{"email":"student@example.com","password":"data260"}' \
http://127.0.0.1:8080/api/auth/login

curl -s -b /tmp/hmcteachlee \
-H "Content-Type: application/json" \
-d '{
  "vulnerabilityTitle":"Postman evidence test report",
  "packageName":"postman-evidence-package",
  "submitterEmail":"student@example.com",
  "description":"This temporary record is used for HW4 API evidence.",
  "category":"Input Validation",
  "isAccepted":true
}' \
http://127.0.0.1:8080/api/reports

{"id":"1","name":"Student Body","email":"student@example.com"}HTTP/1.1 201 Created
date: Thu, 26 Sep 2024 03:27:25 GMT
server: python
content-length: 321
content-type: application/json

{"id":"1889","vulnerabilityTitle":"Postman evidence test report","packageName":"postman-evidence-package","submitterEmail":"student@example.com","description":"This temporary record is used for HW4 API evidence.","category":"Input Validation","isAccepted":true,"submissionDate":"2024-09-26T03:27:26","relatedItems":["1"]}
```

POST create report

GET all reports

GET report by ID

PUT update report

DELETE report

Database and Sessions

Database tables and row counts

[illegible]

Naive page size 50

[illegible]

Naive page size 200

```
(base) sruti@Ashishs-MacBook-Pro:dat206-9004 % curl -i -b /tmp/cookies \
  http://127.0.0.1:8080/api/benchmark/reports/feedback/package_size=0
HTTP/1.1 200 OK
date: Thu, 24 Sep 2026 02:43:22 GMT
server: uvicorn
content-length: 3550
content-type: application/json

{"mode":"fixed","package_size":10,"sql_statements":12,"rows_returned":10,"rows":[{"id":"5003","vulnerability_title":"Updated evidence vulnerability","package_name":"updated-evidence-package","category":"Input Validation","related_ids":[]},{"id":"5004","vulnerability_title":"Seeded vulnerability 2","package_name":"package-2","category":"Dependency Risk","related_ids":["id:305","text:Related benchmark note 18 for vulnerability 5004-0"]},{"id":"5005","vulnerability_title":"Seeded vulnerability 3","package_name":"package-3","category":"Input Validation","related_ids":["id:306","text:Related benchmark note 18 for vulnerability 5005-0"]},{"id":"5006","vulnerability_title":"Seeded vulnerability 4","package_name":"package-4","category":"Dependency Risk","related_ids":[]},{"id":"5007","vulnerability_title":"Seeded vulnerability 5","package_name":"package-5","category":"Input Validation","related_ids":[]},{"id":"5008","vulnerability_title":"Seeded vulnerability 6","package_name":"package-6","category":"Dependency Risk","related_ids":["id:307","text:Related benchmark note 18 for vulnerability 5008-0"]},{"id":"5009","vulnerability_title":"Seeded vulnerability 7","package_name":"package-7","category":"Input Validation","related_ids":[]},{"id":"5010","vulnerability_title":"Seeded vulnerability 8","package_name":"package-8","category":"Dependency Risk","related_ids":["id:172","text:Related benchmark note 18 for vulnerability 5010-0"]},{"id":"5011","vulnerability_title":"Seeded vulnerability 9","package_name":"package-9","category":"Input Validation","related_ids":[]},{"id":"5012","vulnerability_title":"Seeded vulnerability 10","package_name":"package-10","category":"Dependency Risk","related_ids":[]}]}
(base) sruti@Ashishs-MacBook-Pro:dat206-9004 %
```

Fixed page size 10

EXPLAIN before index

EXPLAIN after index

[illegible][illegible]

RAG evaluation results

Complete RAG evaluation

Q5 and Q6 refusals

RAG k sweep

Verification

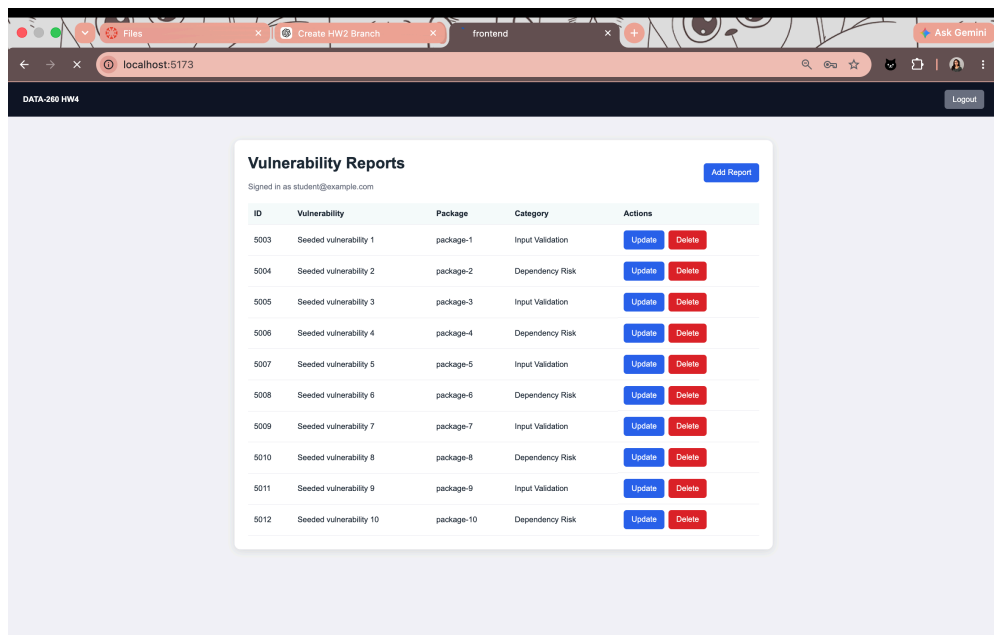
```

data260-9004 --zsh
data260-9004 --zsh
data260-9004 --zsh
data260-9004 -- curl http://localhost:8004/health

"model": "sentence-transformers/all-MiniLM-L6-v2",
"database": "s9004_rel",
"checks": {
  "corpus_has_at_least_5_documents": true,
  "database_has_5000_vulnerabilities": true,
  "database_has_200_related_items": true,
  "n_plus_one_raw_json_valid": true,
  "n_plus_one_has_100_requests": true,
  "n_plus_one_metrics_valid": true,
  "n_plus_one_has_6_metric_groups": true,
  "rag_results_valid": true,
  "rag_evaluation_valid": true,
  "rag_k_sweep_valid": true,
  "rag_printouts_present": true,
  "react_api_present": true,
  "react_app_present": true,
  "database_models_present": true,
  "benchmark_script_present": true,
  "rag_script_present": true,
  "run_log_present": false,
  "metrics_present": false,
  "ai_use_present": false,
  "report_pdf_present": false
},
"status": "incomplete"
}
(base) srusti@Aashiks-MacBook-Pro data260-9004 % cd ~/data260-9004
code reports/hw4/RUN_LOG.txt
(base) srusti@Aashiks-MacBook-Pro data260-9004 % cd ~/data260-9004
code reports/hw4/METRICS.md
(base) srusti@Aashiks-MacBook-Pro data260-9004 % cd ~/data260-9004
code reports/hw4/AI_USE.md
(base) srusti@Aashiks-MacBook-Pro data260-9004 % cd ~/data260-9004
touch reports/hw4/report.md
code reports/hw4/report.md
(base) srusti@Aashiks-MacBook-Pro data260-9004 % uvicorn api:app --app-dir code --reload --port 8004
zsh: command not found: uvicorn
(base) srusti@Aashiks-MacBook-Pro data260-9004 % (base) srusti@Aashiks-MacBook-Pro data260-9004 % uvicorn api:app --app-dir code --reload --port 8004
zsh: command not found: uvicorn
(base) srusti@Aashiks-MacBook-Pro data260-9004 %
zsh: parse error near `srusti@Aashiks-MacBo...'
(base) srusti@Aashiks-MacBook-Pro data260-9004 % uvicorn api:app --app-dir code --reload --port 8004
zsh: command not found: uvicorn
(base) srusti@Aashiks-MacBook-Pro data260-9004 %
code activate data260
(data260) srusti@Aashiks-MacBook-Pro data260-9004 % uvicorn api:app --app-dir code --reload --port 8004
INFO: Will watch for changes in these directories: ['/Users/srusti/data260-9004']
[Error 40] Address already in use
(data260) srusti@Aashiks-MacBook-Pro data260-9004 % uvicorn api:app --app-dir code --reload --port 8004
INFO: Will watch for changes in these directories: ['/Users/srusti/data260-9004']
[Error 40] Address already in use
(data260) srusti@Aashiks-MacBook-Pro data260-9004 %
COMMAND PID USER FD TYPE DEVICE SIZE/OFF NAME
python3.11 24544 srusti 3u IPv4 0x559ebbc2e8a692c3 0t0 TCP localhost:8004 (LISTEN)
python3.11 19951 srusti 3u IPv4 0x559ebbc2e8a692c3 0t0 TCP localhost:8004 (LISTEN)
(data260) srusti@Aashiks-MacBook-Pro data260-9004 % curl http://localhost:8004/health

```

HW4 verification



Frontend dashboard evidence